Chapter 17
Static Electricity

Curiosity Starters

- Why do you sometimes get a small “electric shock” when you touch a metal door handle?
- Why does hair stand up when you rub a balloon on it?
- Why do fuel tankers need metal chains dragging on the ground?
- How does a photocopier use static to print images?

Charging by Friction

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- When two **insulators** are rubbed together, **electrons are transferred** from one surface to the other.
- The object that **loses electrons** becomes **positively charged**.
- The object that **gains electrons** becomes **negatively charged**.
- Charges are always due to movement of **electrons**, never protons (they are fixed inside nuclei).

Example:

- Rubbing a balloon on hair → electrons move from hair to balloon.
- Balloon becomes negatively charged, hair becomes positively charged → hair stands up because like charges repel.

Charging and Discharging

- Static electricity is the result of **electrostatic charges** being built up and held by **insulators**. Since **electrical insulators** prevent electrons from moving freely, the charges remain where they were deposited.
- **Neutral objects** have equal numbers of positive and negative charges. When an object gains or loses electrons, it becomes charged.
- **Electrostatic charge** can attract or repel other charges depending on polarity.

Key Particles and Charges:

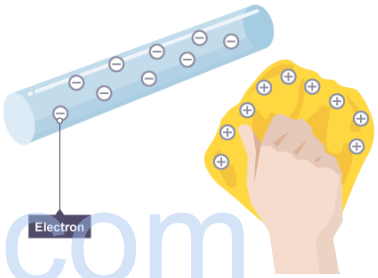
- **Electrons** carry a **negative charge** (-1.6×10^{-19} C).
- **Protons**, found in the nucleus, carry a **positive charge** ($+1.6 \times 10^{-19}$ C).
- When an object becomes negatively charged, it has **more electrons than protons**.
- When it becomes positively charged, it has **lost electrons**.

Conservation of Charge:

- Total charge is conserved.
- One object cannot become negatively charged without the other becoming equally positively charged.

Example:

- Rub a polythene rod with a dry cloth:
  - **Polythene gains electrons** → negatively charged.
  - **Cloth loses electrons** → positively charged.



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Discharging Methods:

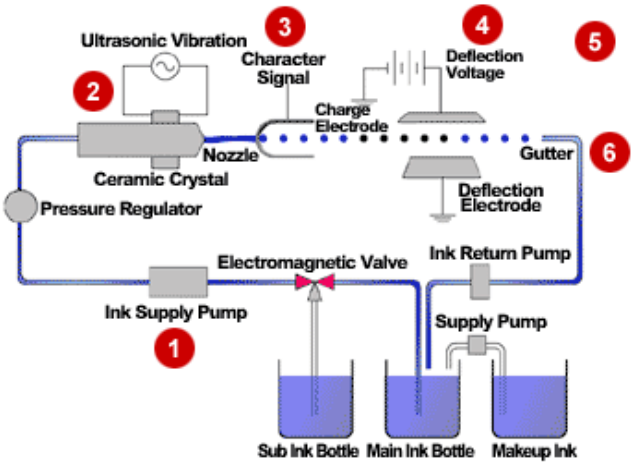
- Bringing the charged object into contact with a conductor connected to the earth (grounding).
- Air breakdown (sparks) when the potential difference is large enough.

Real-Life Examples of Static Electricity:

1. Photocopiers and Laser Printers

How it works:
Static electricity is used to attract toner (ink powder) to specific areas of paper. A laser draws an image on a drum by neutralizing charges. Toner sticks to the charged areas and is then transferred to paper.

Cambridge Tip: This is a standard exam application question.



2. Understand Lightning

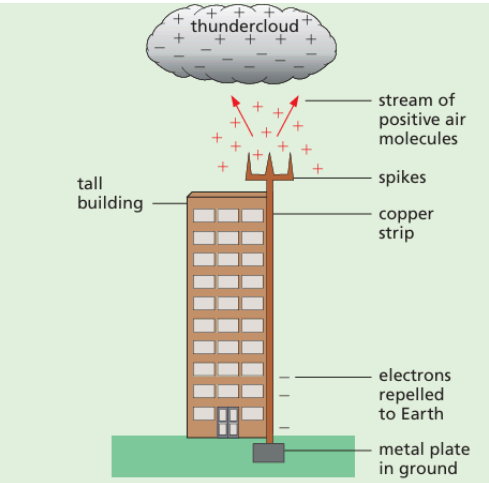


Figure 35.7 Lightning conductor

Clouds become charged due to air turbulence. When the charge difference between clouds and ground becomes large, a spark (lightning) discharges the built-up static electricity.

⚠️ *Why it matters:* Understanding this helps explain why airplanes and fuel stations need proper grounding.

Thought-Provoking Question: Why is it safer to fuel airplanes only after grounding them?

17.2 Explaining Static Electricity

Key Understanding:

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- **Only electrons move** — positive charges are due to electron loss.
- **Insulators retain static charges**, conductors allow charge to flow freely.
- **Friction causes charge separation**, not charge creation.

Important Misconception Clarified:

- Both objects in friction do **not become negatively charged** — one becomes **negative** and the other **positive**. Electrons are transferred from one to the other, conserving total charge.

Charge on Insulators vs Conductors:

- On **insulators**, charge stays in the region where it was transferred.
- On **conductors**, charge spreads out and can escape unless insulated.

Example:

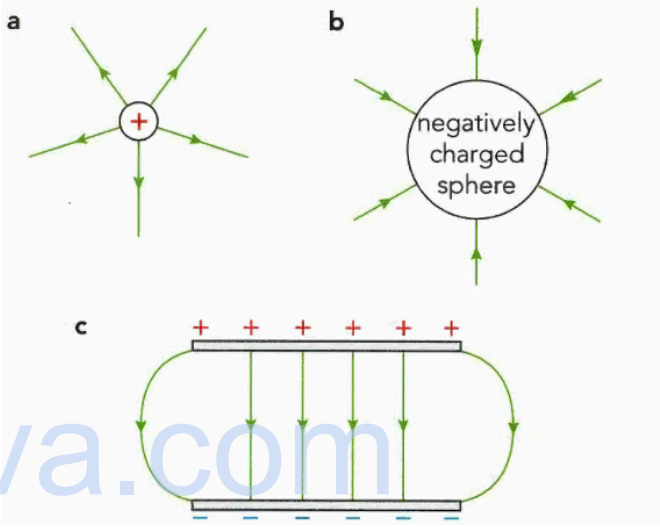
- If you hold the cloth directly, the charge may leak through your body to the ground, making the charge difficult to observe.

Real-Life Link: Shocks from car door handles after sliding off the seat are caused by static buildup discharging into metal.

17.3 Electric Fields

Electric Field:

- A region in which a **charged object experiences a force**.
- **Direction of field lines:** From **positive to negative**.



Equipotential Lines:

- Lines where potential is the same.
- Always **perpendicular to field lines**.

Field Patterns to Learn:

- **Single point charge**: Radial lines.
- **Between two plates**: Uniform field.

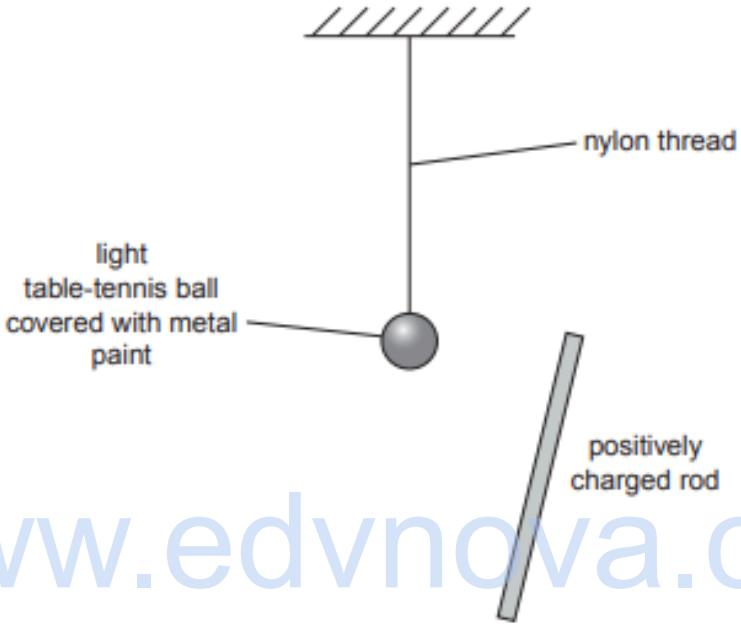
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An Insight: In a **uniform electric field**, a charged particle experiences **constant force**, leading to **uniform acceleration** — similar to gravity.

- **Law of Charges:** Like charges repel, unlike charges attract.
- **Methods of Discharge:** Discharge can occur through sparks, grounding, or air ionization.

Past paper questions

1. A plastic rod is rubbed with a cloth and becomes positively charged. After charging, the rod is held close to the suspended table-tennis ball shown in Fig. 9.1. The table-tennis ball is covered with metal paint and is uncharged.



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Fig. 9.1

(a) Describe what happens to the charges in the metal paint on the ball as the positively charged rod is brought close to the ball.

.....

.....

..... [1]

(b) The ball is attracted towards the charged rod.

Explain why this happens, given that the ball is uncharged.

.....

.....

..... [2]

(c) State the unit in which electric charge is measured.

..... [1]

[Total: 4]

2. **(a)** **(i)** State what is meant by the *direction* of an electric field.

.....

.....[1]

(ii) Fig. 9.1 shows a pair of oppositely-charged horizontal metal plates with the top plate positive.

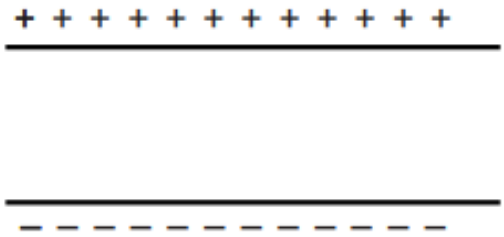


Fig. 9.1

The electric field between the plates in Fig. 9.1 is uniform.

Draw lines on Fig. 9.1 to represent this uniform field. Add arrows to these lines to show the direction of the field. [3]

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(b) Fig. 9.2 shows a very small negatively-charged oil drop in the air between a pair of oppositely charged horizontal metal plates. The oil drop does not move up or down.

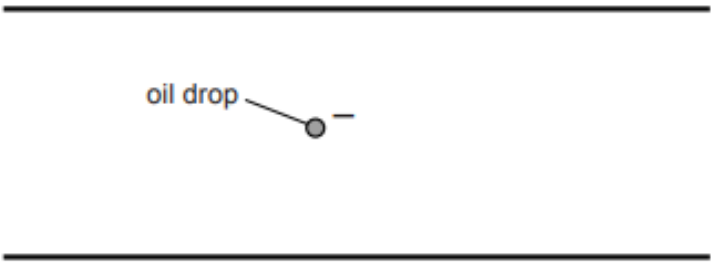


Fig. 9.2

(i) Suggest, in terms of forces, why the oil drop does not move up or down.

.....

.....[2]

(ii) Without losing any of its charge, the oil drop begins to evaporate.

State and explain what happens to the oil drop.

.....

.....[2]

[Total: 8]

3. Fig. 10.1 shows two parallel conducting plates connected to a very high voltage supply.

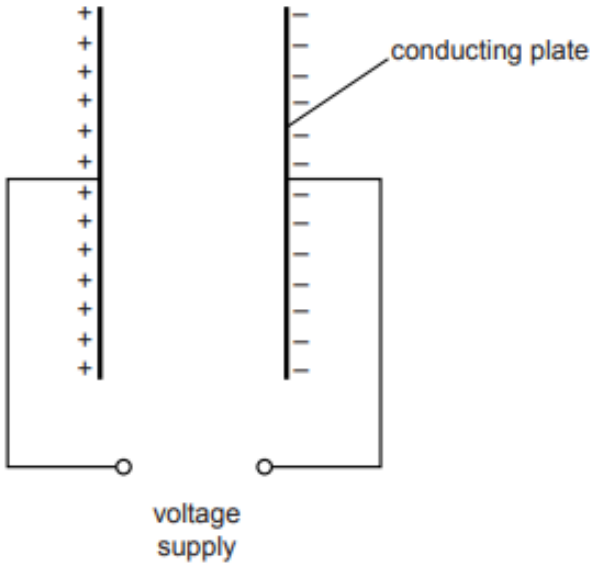


Fig. 10.1

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The left-hand plate is positively charged and the right-hand plate is negatively charged.

(a) On Fig. 10.1, draw the electric field pattern produced between the charged plates. Use arrows to show the direction of the field. [2]

(b) A light, conducting ball is suspended by an insulating string. Fig. 10.2 shows the ball in the middle of the gap between the plates.

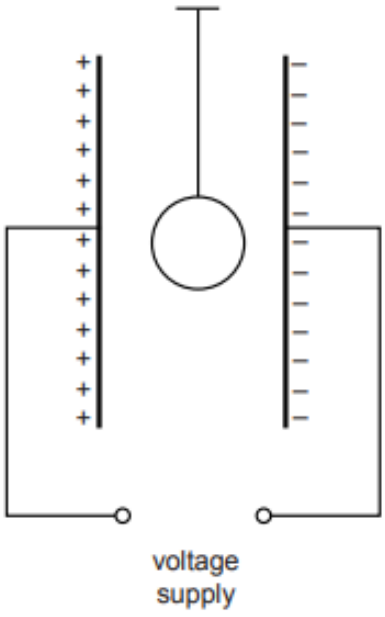


Fig. 10.2

On Fig. 10.2, show the distribution of charge on the ball. [2]

(c) The ball is displaced to the left and then oscillates backwards and forwards between the two plates.

The ball touches a plate once every 0.05 s. Every time it touches a plate, a charge of 2.8×10^{-8} C (0.000 000 028 C) is transferred.

Calculate the average current produced by the repeated transfer of charge.

current = ... [2]

[Total: 6]

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